

Excel for DOA

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❖ Short Cuts

Double Click on Ribbon name to close Ribbon

Enter: To move down to column to another cell

Ctrl + Shift + Arrow key (↓,→,↑,←): To select all content in particular column

Ctrl + Shift + U: To Expand Formula Input box

1. Spread Sheets Intro

1.1 Worksheets



Sheet / Worksheet

(Sheets Inside a workbook)

Formulas

=A1&"-"&E1 → This Will Add Hyphen (-) between data of A1 and E1 cells (Quotes are mandatory)

Ranges – Collection of Cells

While selecting you may notice this inside Name Box:

5R X 2C → Which means you have selected 5 Rows and 2 Columns

To Copy:

AKA (Also Known as) Dynamic Ranges

=A1:G6 → It will copy content from A1 cell till G6 cell to current cell

To Copy Whole Column:

=A:A → Copy All the content of Column A

To Copy Whole Rows:

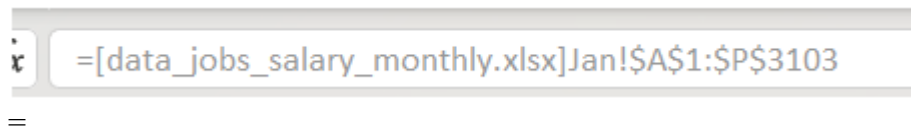
=5:6 → Copy Row 5 and 6

1.2 Workbook

While working with workbooks

3 methods to copy the content of another workbook into our current one:

1) Formula method



We just need to select the cells in the other workbook – It will automatically take this formulas.

Shortcuts:

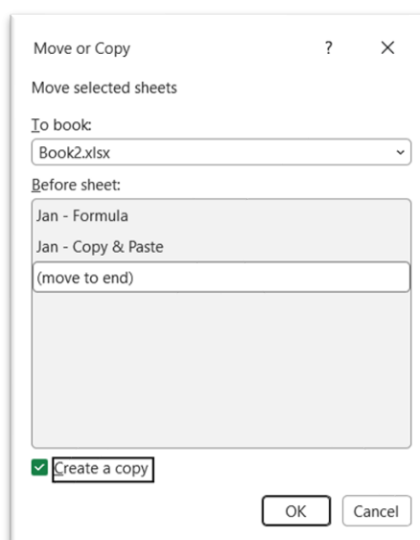
- To Select All rows: Ctrl + Shift + → (Basically to move to end to Right – If standing at left)
- To select All columns: Ctrl + Shift + ↓

2) Copy & Paste Method

Simply go to Data Workbook, Copy all the content and Paste it in a new Workbook

3) Move Or Copy

Just go to Data Workbook, Right Click – Move or copy - “” Create Copy



1.3 File Menu / Back Stage View / Ribbon Menu

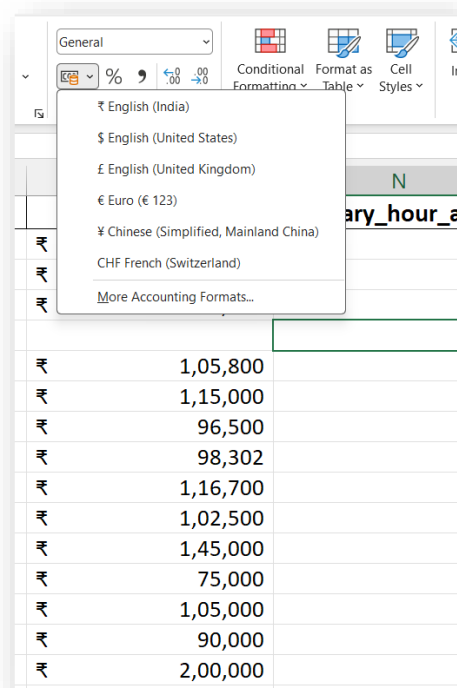
1)

Inside a large data set If we have “#####” showing just double click to expand the size of the column and you will see the actual content of that. 🙄

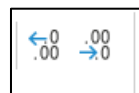
2) If all the columns are collapsed and you want to open all of the columns then:

➔ Select all (full / complete) columns and just double click on the last one to open, It will open all the columns that are selected.

3) To assign a currency symbol to salaries



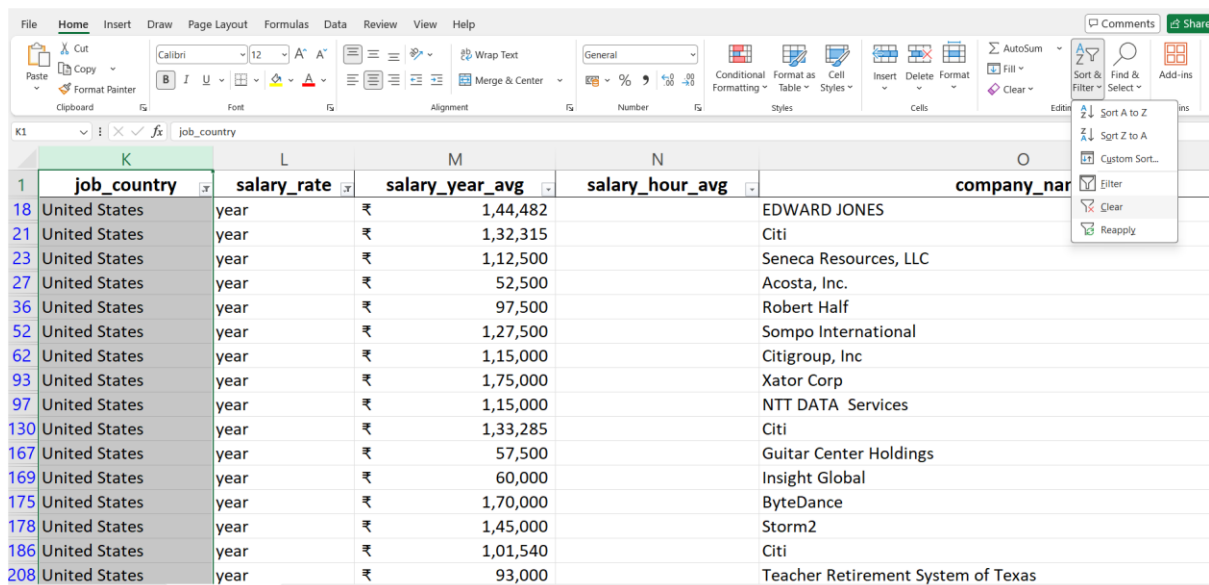
And can also remove the decimals at suffix using



4) To add Filter Options to columns

- Select columns
- Click on Filter (Funnel)

Similarly, If you want to clear all the applied filters then:



Select Any Column and click on Clear

2. Formulas And Functions

2.1 Formulas Intro

- Note that In comparison operator for EQUALS we have only one (=) instead of (==)

- If we want to fix our cell in formula then we can use '\$' sign before the row and column, Eg:

Job Title	Experience (Years)	Annual Salary (\$USD)	Bonus Max (\$USD)	Total Salary (+)	Meets Experience	Meets Salary
Data Scientist	5	120000	10000	130000	=C3<=\$C\$15	
Machine Learning Engineer	4	135000	12000	147000	TRUE	TRUE
Data Analyst	2	75000	5000	80000	TRUE	FALSE
Business Intelligence Dev	6	110000	8000	118000	FALSE	TRUE
Data Engineer	3	125000	11000	136000	TRUE	TRUE
Statistician	7	90000	7000	97000	FALSE	TRUE
Data Architect	10	150000	15000	165000	FALSE	TRUE
Research Scientist	8	130000	13000	143000	FALSE	TRUE
AI Specialist	3	140000	14000	154000	TRUE	TRUE
Big Data Engineer	5	115000	9000	124000	TRUE	TRUE
Job Goals:						
Experience (<=)	5					
Total Salary (>=)	90000					

If we don't want to manually type '\$', then just press "F4" while being cursor on that, It will automatically add '\$' before row and column.

When we do this **\$C\$15**, That is Fixed Reference.

We can even have mixed reference,

\$C15 → Will lock C column but will iterate through rows

C\$15 → Will lock 15 row but will iterate through columns

NOTE:

If You want to Write function as a String, You can add one Quote (') ahead of that and insert it!

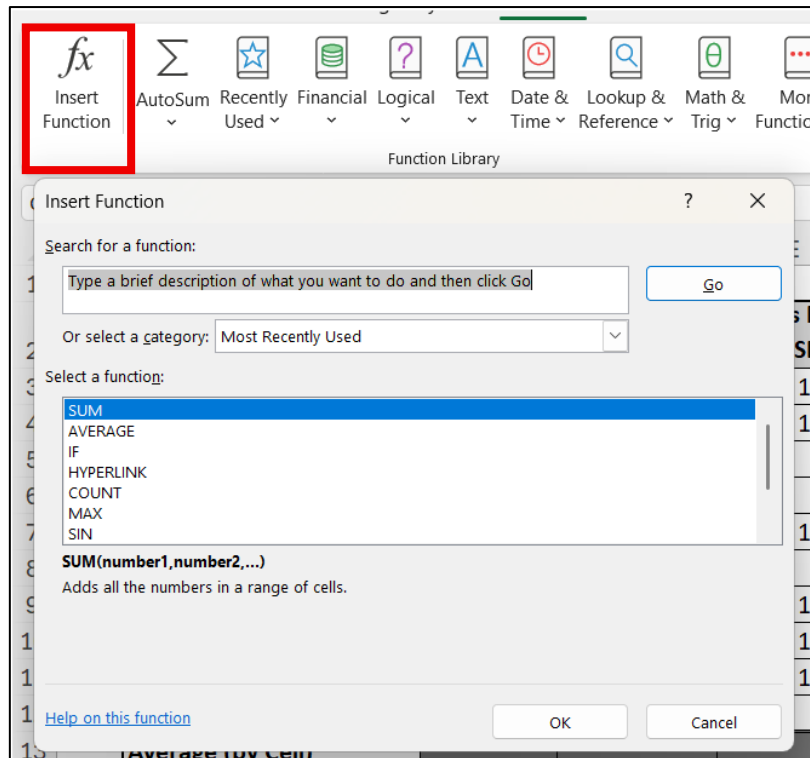
```
'=(A2:A32673=S1)*(M2:M32673<>"")
```

2.2 Functions Intro

=AVERAGE()

=AND()

Also If we are not able to recall the function name, Then we can use this



If you face error like this: COUNTIF()

	C	D	E	L	M	O	P
	Experience (Years)	Annual Salary (\$USD)	Bonus Max (\$USD)				
	5	120000	10000				
g Engineer	4	135000	12000				
	2	75000	5000				
ence Dev	6	110000	8000				
	3	125000	11000				
	7	90000	7000				
	10	150000	15000				
st	8	130000	13000				
	3	140000	14000				
er	5	115000	9000				
)							
GE())	5.3	119000	10400				
	10	10	10				
unt	=COUNTIF(C3:C12,<=5)						
unt							

Microsoft Excel

There's a problem with this formula.

⚠ Not trying to type a formula?
When the first character is an equal ("=") or minus ("-") sign, Excel thinks it's a formula:

- you type: =1+1, cell shows: 2

To get around this, type an apostrophe (') first:

- you type: '=1+1, cell shows: =1+1

OK

Then to solve this, we need to wrap our conditional operators around “ ” (Quotes),

If we want to Refer a cell inside complicated functions like COUNTIF(), We need to use “&”

	C	D	E	L	M	O	P	Q	R	S	T
	Experience (Years)	Annual Salary (\$USD)	Bonus Max (\$USD)	Meets Experience	Meets Salary	Meets Both - Formula Only	Meets Both - AND()		Job Goals:		
	5	120000	10000	TRUE	TRUE	TRUE	TRUE		Experience (<=)	5	
nee	4	135000	12000	FALSE	TRUE	FALSE	FALSE		Total Salary (>=)	90000	
	2	75000	5000	FALSE	FALSE	FALSE	FALSE				
ev	6	110000	8000	TRUE	TRUE	TRUE	TRUE				
	3	125000	11000	FALSE	TRUE	FALSE	FALSE				
	7	90000	7000	TRUE	TRUE	TRUE	TRUE				
	10	150000	15000	TRUE	TRUE	TRUE	TRUE				
	8	130000	13000	TRUE	TRUE	TRUE	TRUE				
	3	140000	14000	FALSE	TRUE	FALSE	FALSE				
	5	115000	9000	TRUE	TRUE	TRUE	TRUE				
	5.3	119000	10400								
	10	10	10								
	6	9									
	=COUNTIF(\$C\$3:\$C\$12,"<="&\$S3)										
	COUNTIF(range, criteria)										

2.3 Logical Functions

IF() → IF(Cell_location, “What to print If true”, “What to print if False”)

→ Nested IF:

IF(Cell_location, “What to print If true”, IF(Cell_location, “What to print If true”, “What to print if False”))

→ IFS:

- Multiple If conditions

- IFS(“Condition1”, “Value if true1”, “Condition2”, “Value if true2”, “Condition3”, “Value if true3”)

2.4 Math Functions

Function	What it Counts
COUNT()	Counts only cells containing numbers
COUNTA()	Counts all non-empty cells (numbers, text, dates, logical values, errors, etc.)

Function	Number of Conditions	Purpose
COUNTIF	1 condition	Counts cells that satisfy one criteria.
COUNTIFS	Multiple conditions	Counts cells that satisfy all the specified criteria.

→SUMIF:

SUMIF(*Range to check condition*, *To match with?*, *Range for sum*)

→ SUMIFS:

SUMIFS(*Range for sum*, *Range to check condition1*, *To match with1?*, *Range to check condition2*, *To match with2?*)

→ MINIFS:

→ MAXIFS:

2.5 Statistical Functions

Median(), Min() ,Max()

STDEV → Standard Deviation

→ Quartile.INC(array, quartile)

→ Quartile.EXC(array, quartile)

➔ Quartile Inclusive and Exclusive both provides same result, just the difference between them is Quartile.EXC() – excludes quartile 0 and 4 that MIN and MAX.

Also:

quartile 0 = gives similar value to MIN()

quartile 4 = gives similar value to MAX()

→ =MODE()

In Excel, =MODE() is used to find the **most frequently occurring value** in a range of numbers.

Example

If cells A1:A6 contain:

	A
1	10
2	20
3	20
4	30
5	20
6	40

=MODE(A1:A6)

RESULT: 20

→ RANK():

We can also rank the columns based on (let's a salary)

RANK(current cell's rank reference, All the values (Array - Range))

2.6 Array Formulas

Modern (Dynamic) Arrays | **VS** | Classic(Old) Arrays

Modern (Dynamic) Arrays:

- They are modern / new arrays
- It provides ease as compare to classic Arrays
- Has simple formula
- E.g. =B2:B6
- Just need to press “Enter” after writing formula
- When you click on one of the cell in it - borders becomes have some shadows (in excel)
- Directly write formula

Classic (Old) Arrays:

- They are old arrays
- They are more complex
- Has complex formula
- E.g. {=A2:A6*B2:B6}
- We need to press Ctrl + Shift + Enter after writing formula
- No shadows around border of table
- First we need to select the cells and then need to write formula

→ =UNIQUE()

It will return all unique values once – It is like group by of SQL – just difference is it will return the values of group by (Distinct values)

→ =SORT(range)

One term I learnt recently is Spill range.

So specify here spill range

→ #SPILL! – Error

When We have Spilled range, and if we try to change a single cell, then it will give us SPILL Error - Because we are manually modifying the cell and destroying the consistency.

#SPILL! is an Excel error that occurs when a formula wants to return multiple values (a spilled array), but Excel cannot place the results into the required cells.

→MEDIANIF():

By default Excel does not provide MEDIANIF(), So to have that We need to put IF condition inside MEDIAN(IF())

E.g.

```
=MEDIAN(IF(($A$2:$A$32673=$AA2)*($M$2:$M$32673<>""),$M$2:$M$32673))
```

This way we can work with *ANY* function as having the IF functionality, by providing IF() inside that function.

■ If we want to convert TRUE / FALSE value to Integer, we can do

-(TRUE) → -1 (Negates one time)

--(TRUE) → 1 (Negates two time – means - - = +)

→ TEXT() function

- It is for formatting

2.7 Lookup Functions

→ VLOOKUP()

VLOOKUP = Vertical Lookup

It searches for a value in the **first column** of a table and returns a value from another column in the **same row**.

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
--

Parameters

- **lookup_value** → Value to search for.
- **table_array** → Range containing the data.
- **col_index_num** → Column number (starting from 1) from which to return the result.
- **range_lookup**
 - FALSE = Exact match (most commonly used)
 - TRUE = Approximate match

In VLOOKUP(), **col_index_num** means:

The column number from the selected table range from which you want Excel to return a value.

→ Generally we use “FALSE” in *range_lookup* because, by default they expect the values to be sorted but as because in our case they are not sorted, hence we provide FALSE means do Exact match.

→ HLOOKUP():

HLOOKUP = Horizontal Lookup

It searches for a value in the **first row** of a table and returns a value from another row in the **same column**.

=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])
--

→ MATCH():

The **MATCH()** function returns the **position** of a value within a range, not the value itself.

```
=MATCH(lookup_value, lookup_array, [match_type])
```

Parameters

- **lookup_value** → The value you want to find.
- **lookup_array** → The range to search in.
- **match_type** (optional):
 - 0 → Exact match (most commonly used)
 - 1 → Largest value $\leq$ lookup_value (ascending order required)
 - -1 → Smallest value $\geq$ lookup_value (descending order required)

➔ **XLOOKUP()**:

XLOOKUP() is a modern lookup function in Excel that replaces both VLOOKUP() and HLOOKUP(). It can search vertically or horizontally and is more flexible.

```
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])
```

Argument	Description
lookup_value	Value you want to find
lookup_array	Range where Excel searches for the value
return_array	Range from which Excel returns the result
if_not_found	Optional message if value is not found
match_mode	Optional match type
search_mode	Optional search direction

2.8 TEXT Functions

➔ **TEXTJOIN()**:

TEXTJOIN() combines text from multiple cells into a single string and lets you specify a separator (delimiter).

=TEXTJOIN(delimiter, ignore_empty, text1, [text2], ...)

Argument	Description
delimiter	Character(s) between values (e.g., ", ", " - ", " ")
ignore_empty	TRUE to skip blank cells, FALSE to include them
text1, text2...	Cells, ranges, or text to join

→ TEXTSPLIT():

TEXTSPLIT() is an Excel function used to **split text into multiple cells** based on a delimiter (separator).

=TEXTSPLIT(text, col delimiter, [row delimiter])

Parameters

- **text** → Text to split
- **col_delimiter** → Character(s) where text should be split into columns
- **row_delimiter** (*optional*) → Character(s) where text should be split into rows

→ RIGHT():

RIGHT() Excel function is used to extract characters from the **right side** of a text string. (i.e. from the end of the text string)

=RIGHT(text, [num chars])

Parameters

- **text** → The text or cell reference.
- **num_chars** (*optional*) → Number of characters to return from the right. Default is 1.

NOTE: In Excel String Indexing starts from 1.

→ **FIND()**:

Finds the position of a character or text string within another text string.

<code>=FIND(find_text, within_text, [start_num])</code>

→ **LEN()**:

Returns the number of characters in a text string, including spaces.

<code>=LEN(text)</code>

→ **MID()**:

Extracts a specific number of characters from the middle of a text string.

<code>=MID(text, start_num, num_chars)</code>

Eg:

`=MID("Nazil Dhalwala",7,8)`

Result: Dhalwala

Explanation:

- Start from position 7
- Extract 8 characters

→ **TRANSPOSE()**:

TRANSPOSE() in Excel is used to **change the orientation of a range of cells**—it converts **rows into columns** or **columns into rows**.

<code>=TRANSPOSE(array)</code>

➔ UNIQUE():

UNIQUE() in Excel returns **only the distinct (unique) values** from a range or array by removing duplicates.

<code>=UNIQUE(array, [by_col], [exactly_once])</code>

Arguments

- **array** (*required*): The range or array from which you want unique values.
- **by_col** (*optional*):
 - FALSE (default): Compare rows.
 - TRUE: Compare columns.
- **exactly_once** (*optional*):
 - FALSE (default): Returns all distinct values (each value appears once in the result).
 - TRUE: Returns only values that appear **exactly one time** in the source data.

2.9 Date & Time Functions

➔ MONTH():

➔ DAY():

➔ YEAR():

➔ DATE():

➔ TODAY():

➔ DATEDIF("Start date (from)", "to date", "D" [format]):

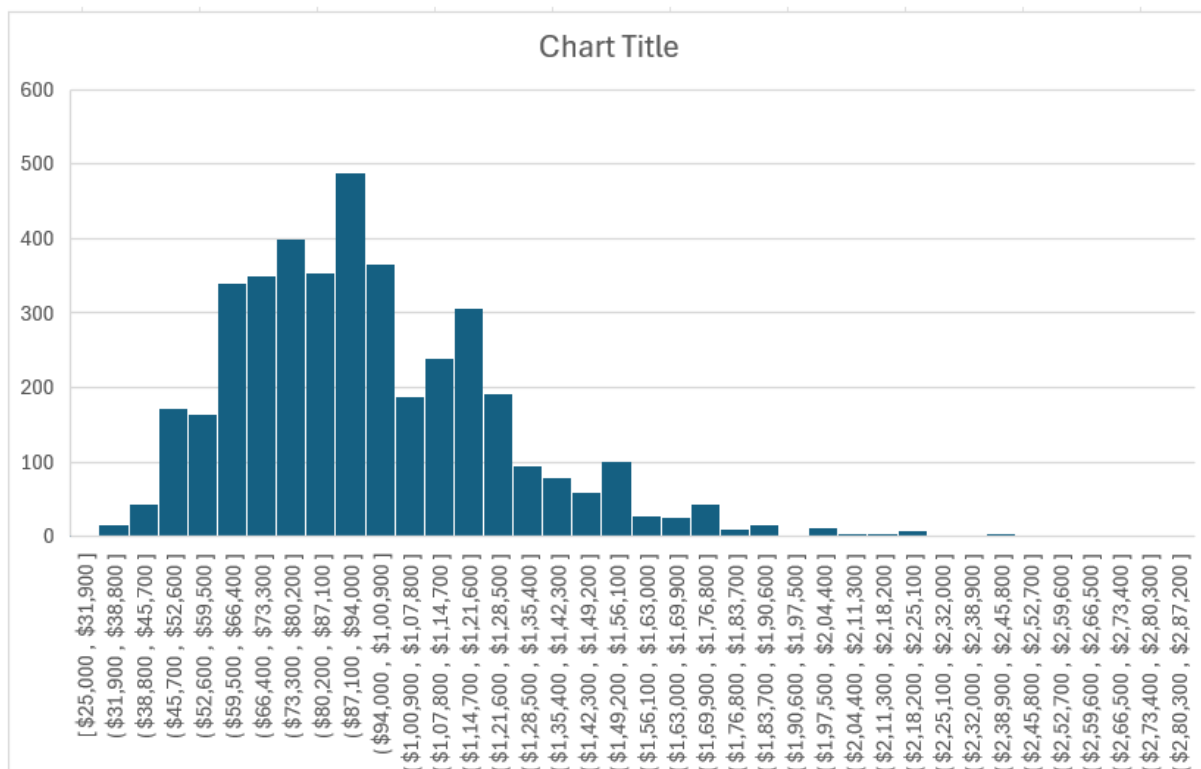
➔ HOUR():

➔ MINUTE():

- ➔ SECOND():
- ➔ TIME():
- ➔ SEQUENCE():

3. Charts And Graphs

3.3 Charts – Statistics



Blue bars are known as “Bins”

We can adjust the size of bins.

3.4 Sparklines

Sparklines are visualization of data within the cell.

	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Sparkline
46	933	856	1031	733	797	645	663	
19	161	117	172	111	95	75	74	
32	806	781	965	730	655	507	530	
59	205	184	234	137	124	99	103	
16	711	543	671	589	522	381	396	
33	228	184	242	157	140	93	104	
35	3044	2665	3315	2457	2333	1800	1870	

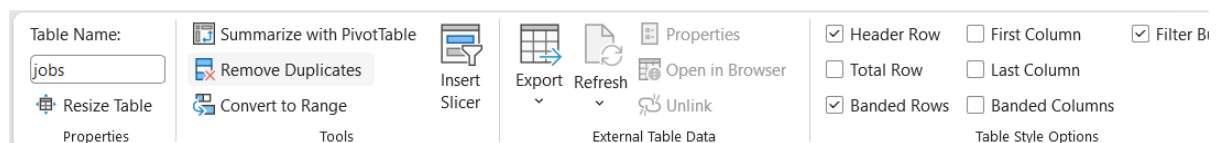
4. Spreadsheets Advanced

4.1 Tables

To create table:

If you already have a dataset with you, keep your cell on 1st cell of table and row - and then go to – Insert > Table

- When you will create a new column just (right) next to your table - Excel will automatically consider that column to table's column and will include newly created column to table
- To copy column –
 - `=[@[column_name]]` (just select the cell – Excel automatically fills the formula)



Keep the table name simple – So that It would be easy for you to refer later on...

To refer the created table, we can use

```
=table_name
OR
=table_name[#All]
```

OR
 =table_name[#Headers]
 OR
 =table_name[column_name] ➔ To access particular column
 Etc.....

=jobs												
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
job_title_s	job_posted	job_location	job_count	company	job_via	job_sched	salary_yea	job_work	job_skills	salary_year_avg	copy	
Senior Data	44996.5	Aliso Viejo	United States	Adroit Wor	via ZipRec	Full-time	135000	FALSE	['sql', 'pyth	135000		
Data Analyst	45097.9	Anywhere	Malta	Jobmatchi	via LinkedIn	Contractor	35000	TRUE	['sql', 'r', 'p	35000		
Data Scientist	45156.3	Cupertino, United States		Apple	via Snagaj	Full-time	0	FALSE	['sql', 'pyth	0		
Senior Data	45147.8	Rochester, Sudan		Securian F	via Snagaj	Full-time	0	FALSE	['sql', 'aws'	0		
Data Analyst	44949.9	Atlanta, Georgia	United States	The E Group	via ZipRec	Full-time	0	FALSE	['sql', 'pyth	0		
Data Scientist	45029.9	Sunnyvale	United States	Walmart	via Indeed	Full-time	195000	FALSE	['python', '	195000		
Senior Data	45091.5	Pasadena, United States		Harnham	via Indeed	Full-time	165000	FALSE	['python', '	165000		
Data Scientist	45175.5	Houston, Texas	United States	Stripe	via IT JobSi	Full-time	281451	FALSE	['python', '	281451		
Data Scientist	45117.8	Johannesburg	South Africa	Standard E	via Ai-Jobs	Full-time	70000	FALSE	['sql', 'pyth	70000		
Data Engineer	44938.6	Berlin, Germany		Vattenfall	via Ai-Jobs	Full-time	98301.5	FALSE	['python', '	98301.5		
Data Engineer	44985.4	Amsterdam	Netherlands	Devoteam	via Ai-Jobs	Full-time	89100	FALSE	['sql', 'gcp'	89100		
Business Analyst	45063.8	Dallas, Texas	United States	DISYS	via Indeed	Contractor	0	FALSE	['sql', 'html	0		
Data Scientist	45017	Broomfield	Sudan	Datalab US	via Indeed	Full-time	87500	FALSE	['sql', 'r', 'p	87500		
Data Scientist	45169.6	Anywhere	Sudan	Fortira Inc.	via LinkedIn	Contractor	0	TRUE	['sql', 'sas',	0		
Business Analyst	45000.5	Palo Alto, California	United States	Denodo Te	via Ladder	Full-time	90000	FALSE	['power bi'	90000		

- We can include “Total” row:

☒ Header Row	☐ First Column	☒ Filter Button
☒ Total Row	☐ Last Column	
☒ Banded Rows	☐ Banded Columns	
Table Style Options		

- When we will include Total row, then Total row uses special formula / function
 =SUBTOTAL(func_number,[range])
 =AGGREGATE(func_number,options,[range])

Luke suggests and recommends to use this formulas in Total rows.

❖ Inserting Slicers:

Insert Draw Page Layout Formulas Data Review View Help **Table Design**

Summarize with PivotTable
Remove Duplicates
Convert to Range

Insert Slicer

Export
Refresh

Properties
Open in Browser
Unlink

☒ Header Row
☐ Total Row
☒ Banded Rows

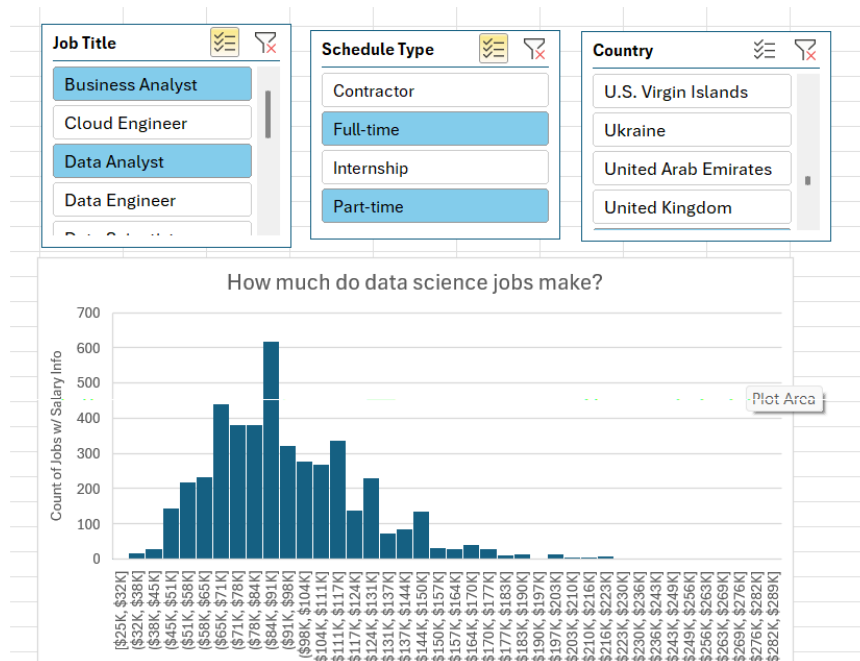
☐ First Column
☐ Last Column
☐ Banded Columns

Tools External Table Data Table Style

fx 90000

	B	C	D	E	F	G
ort	job_schedule_type	job_country	salary_year_avg			
	Full-time	United States	\$ 1,49,509			
	Full-time	United States	\$ 1,25,000			
	Full-time	United States	\$ 84,000			
	Full-time	United States	\$ 75,000			
	Full-time	United States	\$ 1,00,000			
	Full-time	United States	\$ 90,000			
	Full-time	United States	\$ 70,000			

Job Title
Business Analyst
Cloud Engineer
Data Analyst
Data Engineer

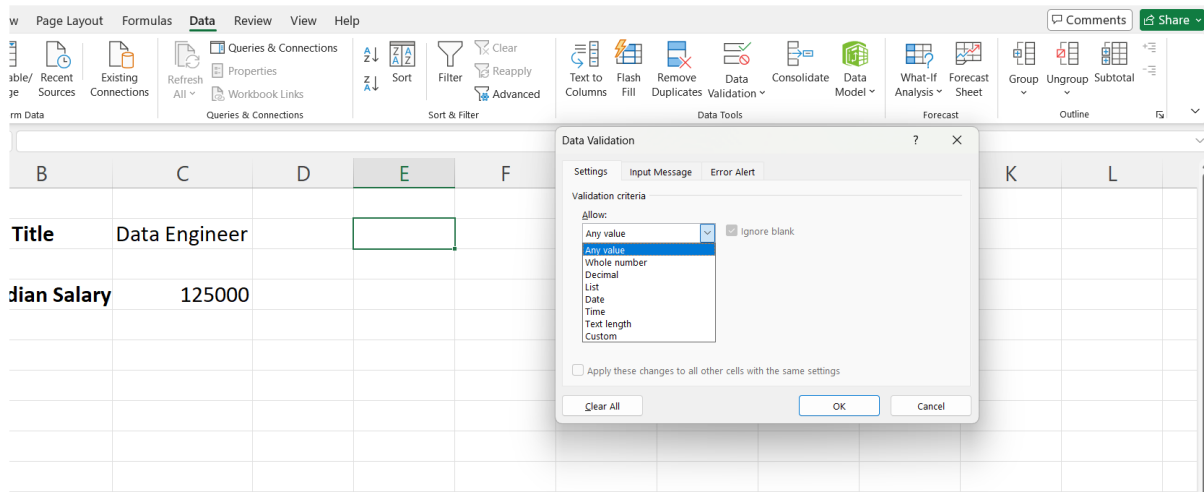


4.3 Collaboration

Data Validation

We can make our cell as List by this:

Data > Data Validation > Allow:List



Note: We can Rename any Excel cell to anything and can access that cell in any sheet.

E.g.

From C2 → title

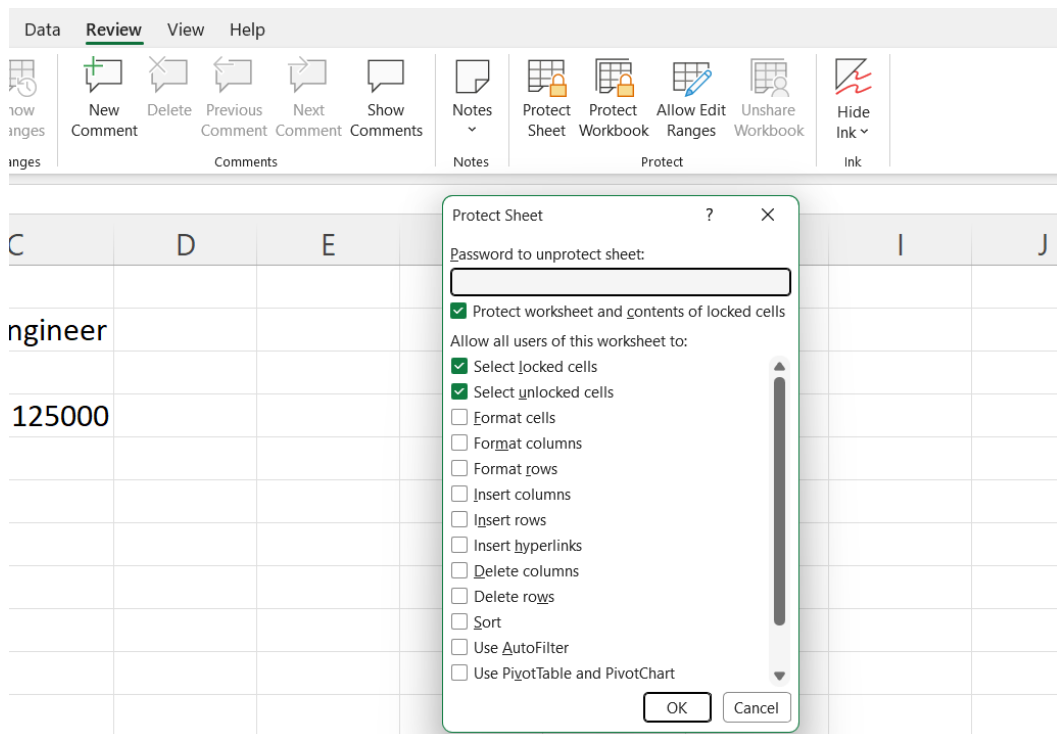
	A	B	C	D
1				
2		Job Title	Data Engineer	
3				
4		Median Salary	125000	

Accessing it:

=VLOOKUP(title,D2#,2,FALSE)				
	D	E	F	G
Senior Data Scientist		155000		
Senior Data Engineer		147500		125000
Data Scientist		127500		

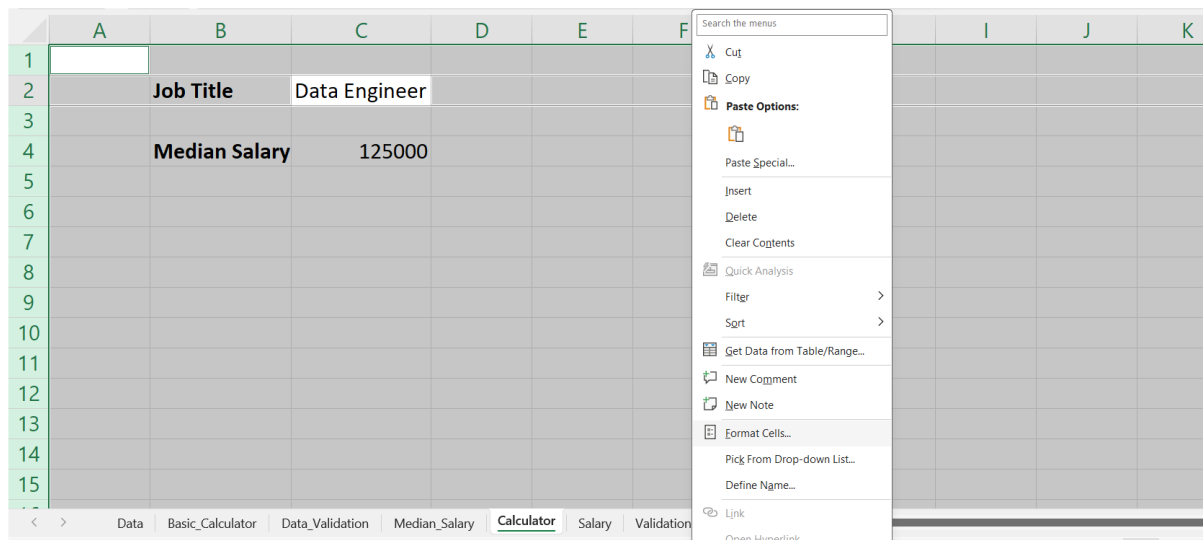
How to Protect Sheets?

Under Review Tab, We have this Protech Sheet, Protect Workbook Option:



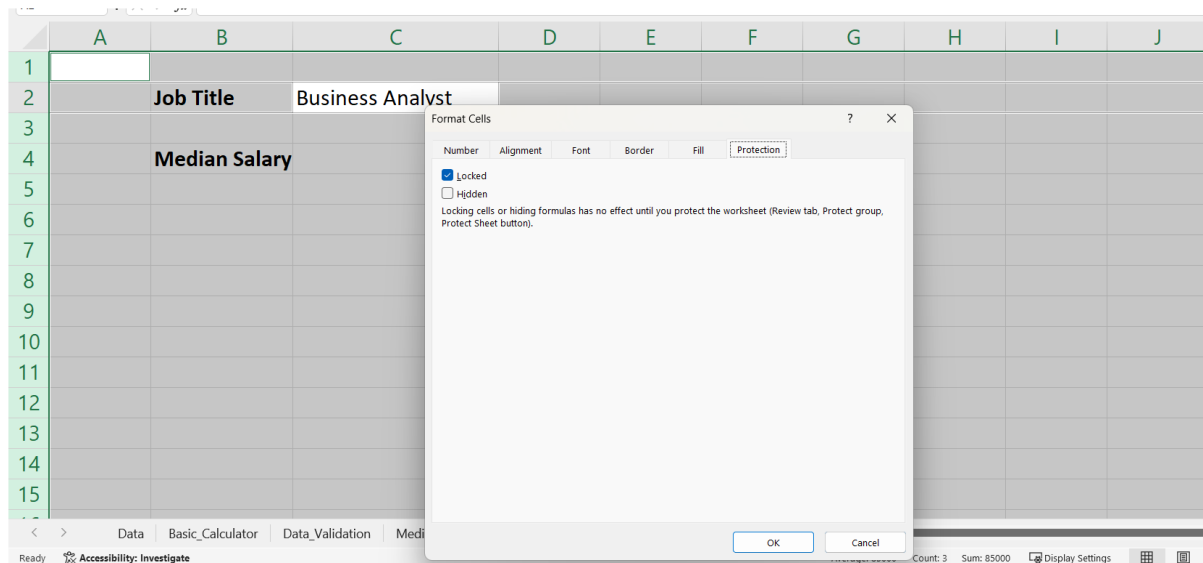
Allowing user to access only List

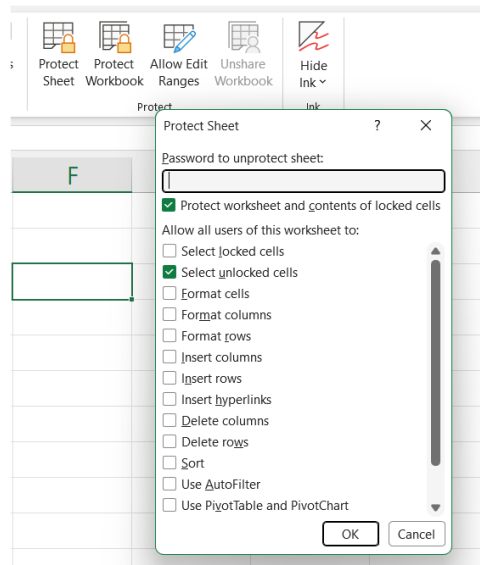
Select all > Right Click > Format Cells



Check on “Locked” and now, All the selected cells are locked.

But Immediately after this, It won't work. To work You have “**Protect Sheet**”.

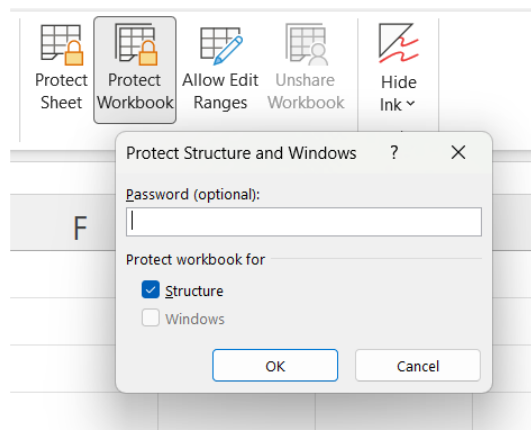




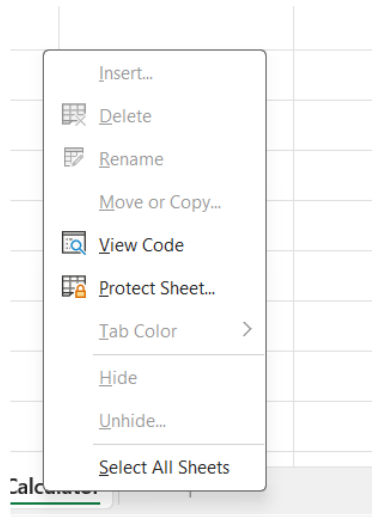
How to protect Workbooks?

This can be useful – If we have multiple workbooks available in sheet, and You want to hide some of workbooks of sheet and don't want user to unhide those hidden workbooks and see them.

For this we can use “**Protect Workbook**”



Now, User can't unhide workbooks



Project 1: Dashboard

→ SEARCH():

The **SEARCH()** function in Microsoft Excel finds the **position of one text string inside another**. It is **not case-sensitive**.

```
=SEARCH(find_text, within_text, [start_num])
```

Arguments:

- **find_text** – The text you want to find.
- **within_text** – The text in which you want to search.
- **start_num** (*optional*) – The position from which to start searching.

→ FILTER():

The **FILTER()** function returns only the rows or columns that meet a condition.

```
=FILTER(array, include, [if_empty])
```

Arguments:

- **array** – Data to filter.
- **include** – Condition (TRUE/FALSE).
- **if_empty** (*optional*) – Value to return if nothing matches.

➔ **NOT()**:

The NOT() function reverses a logical value.

- TRUE → FALSE
- FALSE → TRUE

```
=NOT(logical)
```

➔ **ISNUMBER()**:

The ISNUMBER() function checks whether a value is a number.

```
=ISNUMBER(value)
```

➔ **SUBSTITUTE()**:

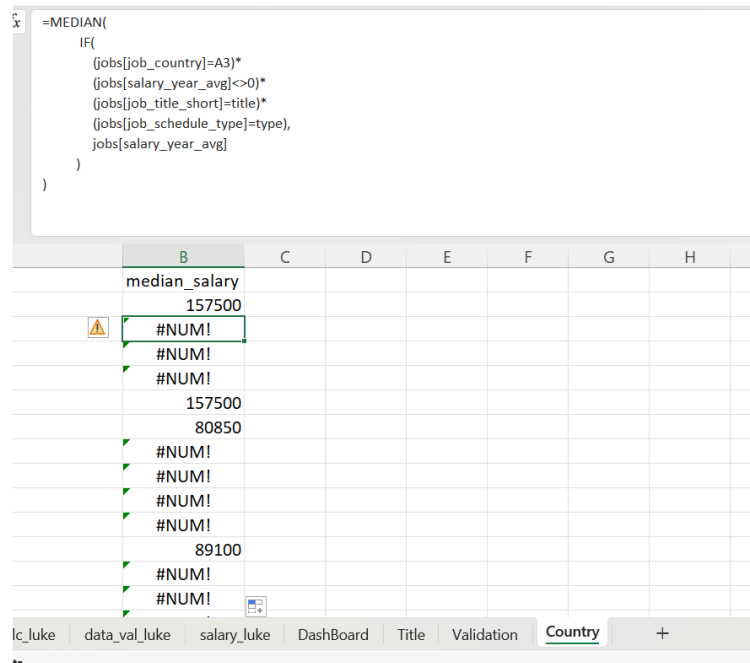
SUBSTITUTE() is an Excel function used to **replace specific text within a string** with different text.

```
SUBSTITUTE(text, old_text, new_text, [instance_num])
```

Parameters

- **text** – The original text.
- **old_text** – The text you want to replace.
- **new_text** – The replacement text.
- **instance_num** (*Optional*) – Specifies which occurrence to replace. If omitted, all occurrences are replaced.

➔ We can do like this for job_schedule_type,



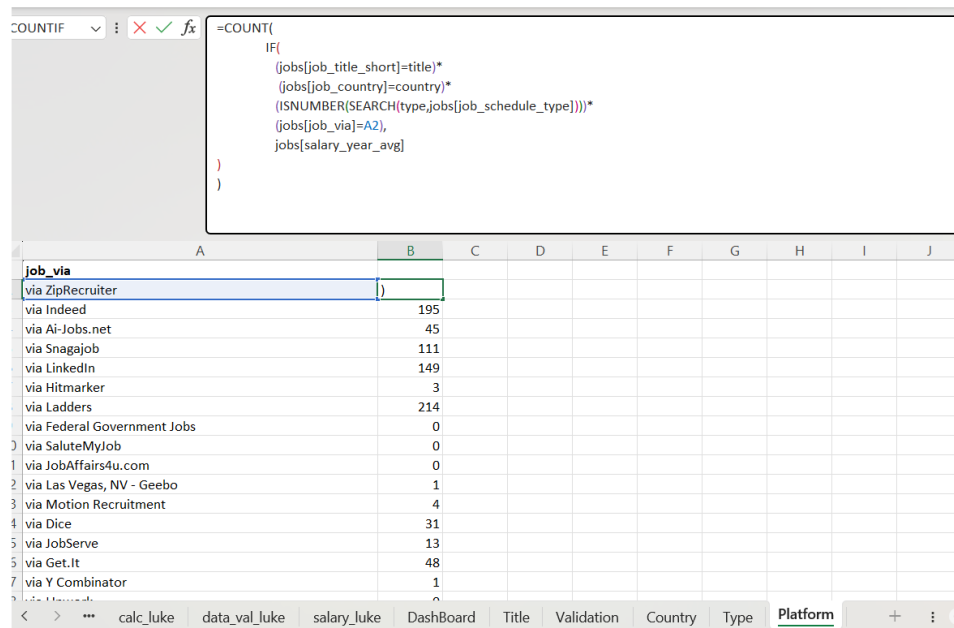
But, Because we had filtered the job_Schedule_type a lot, hence we need to find “type” in to actual job_schedule_type list

M	N	O	P
job_schedule_type		job_schedule_type_sorted	
Full-time		Full-time	
Full-time and Part-time		Contractor	
Contractor		Part-time	
Full-time, Part-time, and Internship		Internship	
Part-time		Temp work	
Contractor and Temp work			
	0		
Full-time and Temp work			
Full-time and Contractor			
Full-time, Contractor, and Temp work			
Full-time, Part-time, and Temp work			
Full-time, Part-time, and Contractor			
Internship			
Full-time and Internship			
Part-time and Internship			
Full-time and Per diem			
Part-time and Contractor			
Temp work and Internship			
Temp work			
Part-time, Contractor, and Temp work			
Internship and Full-time			
Full-time, Part-time, and Per diem			
Part-time and Full-time			
Part-time and Temp work			
Full-time, Temp work, and Internship			
Full-time, Contractor, and Internship			
Full-time, Part-time, and Volunteer			

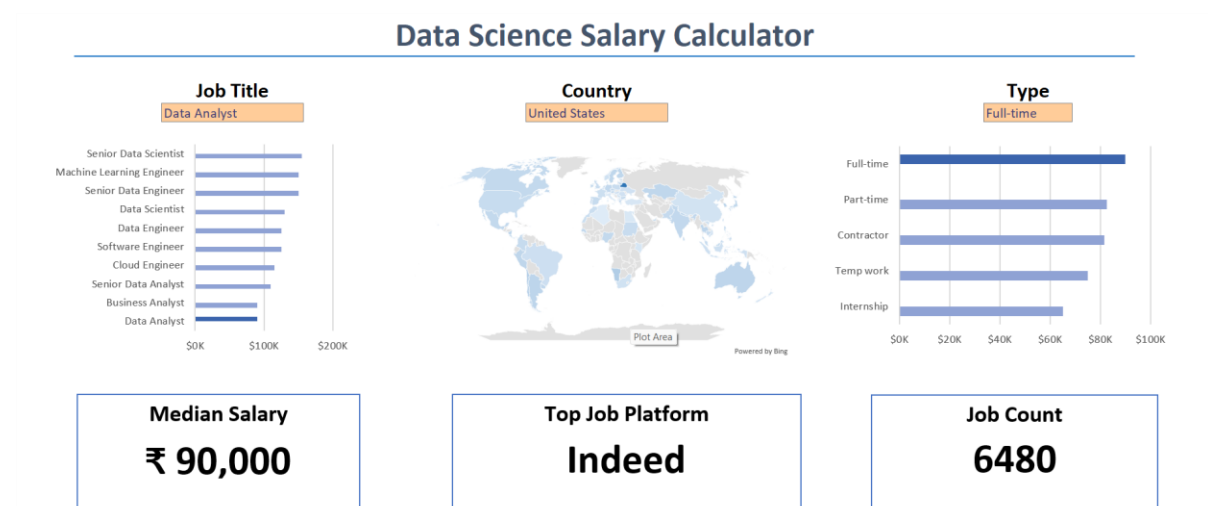
Hence we can SEARCH here – which will return index of that.

➔ We have not included `jobs[salary_year_avg]<>0` condition in job_via / Platform finding workbook, because If we will include it, then It will not count those who are having

salary_year_avg = 0 and but What if some data have salary_year_avg = 0 But have salary_month_avg = <something> ? Hence we are removing that condition.



➔ Final Output of Project:



5. Pivot Tables Intro

To insert pivot table:

File Home **Insert** Draw Page Layout Formulas Data Review View Help

PivotTable Recommended Table Pictures Shapes SmartArt Icons Screenshots Recommended Charts

From Table/Range
From External Data Source
From Data Model

job_title_short

	D	E	F	G	H	I				
	job_via	schedule	ork_from	rch_locati	posted_d	degree_nealt				
2	Senior Dat	Senior Dat	Benningto	via ZipReci	Full-time	FALSE	Sudan	#####	FALSE	T
3	Data Engin	Data Engin	Mesa, AZ	via Indeed	Full-time	FALSE	Georgia	#####	TRUE	T
4	Senior Dat	Supervisor	Alexandria	via ZipReci	Full-time	FALSE	New York,	#####	TRUE	F/
5	Machine L	Machine L	Pittsburgh,	via Ai-Jobs	Full-time	FALSE	Illinois, Un	#####	FALSE	T
6	Data Scien	Data Scien	Irving, TX	via Snagaj	Full-time a	FALSE	Texas, Uni	#####	FALSE	F/
7	Data Engin	Data Engin	Tysons, VA	via Snagaj	Full-time a	FALSE	Sudan	#####	FALSE	F/
8	Data Scien	Sr. Data Er	Anywhere	via Indeed	Full-time	TRUE	Texas, Uni	#####	FALSE	T
9	Senior Dat	Senior Dat	Coplay, PA	via ZipReci	Full-time	FALSE	Illinois, Un	#####	FALSE	F/
10	Senior Dat	Senior Dat	Anywhere	via ZipReci	Full-time	TRUE	California,	#####	TRUE	F/
11	Data Scien	Data Scien	Mountain	via Linked	Full-time	FALSE	California,	#####	FALSE	T
12	Data Scien	University	San Franci	via Snagaj	Full-time a	FALSE	California,	#####	FALSE	F/
13	Data Scien	Staff Data	Los Angele	via Hitmar	Full-time	FALSE	California,	#####	FALSE	T
14	Data Scien	Data Scien	Bellows	via Snagaj	Full-time a	FALSE	California,	#####	FALSE	F/

PivotTable from table or range ? X

Select a table or range

Table/Range: Data!\$A\$1:\$P\$32673

Choose where you want the PivotTable to be placed

☒ New Worksheet

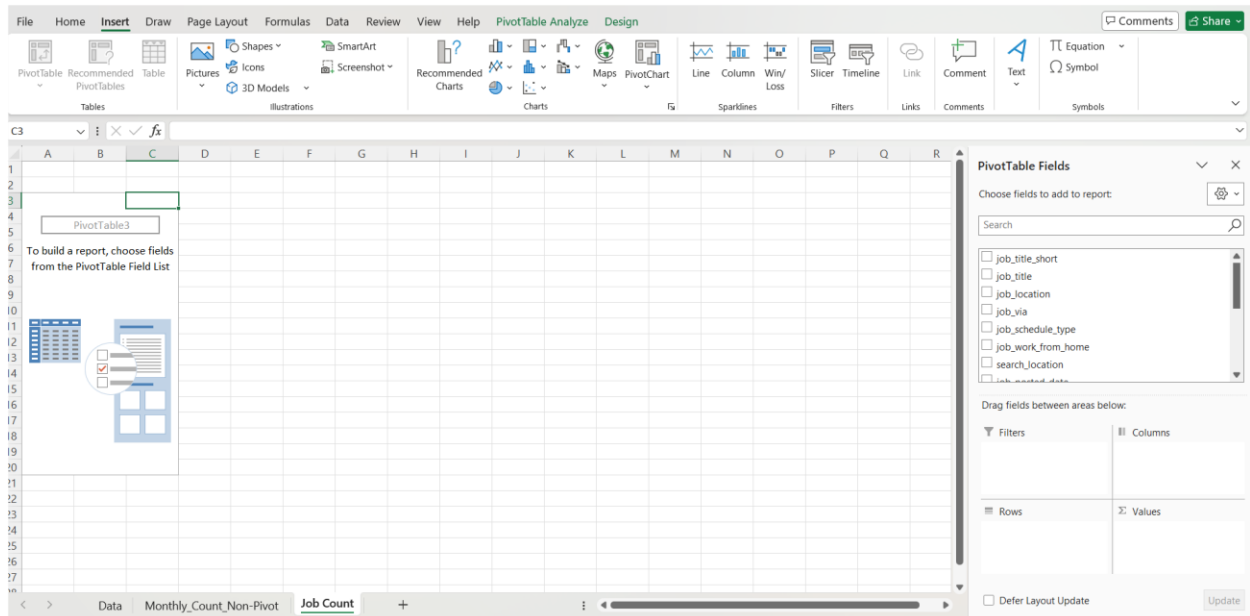
☐ Existing Worksheet

Location:

Choose whether you want to analyze multiple tables

☐ Add this data to the Data Model

OK Cancel



5.2 Pivot Table Advance

This is known as Hierarchy

	A	B
1	Row Labels	
2	Albania	
3	Algeria	
4	Data Analyst	
5	Data Engineer	
6	Argentina	
7	Business Analyst	
8	Cloud Engineer	
9	Data Analyst	
10	Data Engineer	
11	Data Scientist	
12	Machine Learning Engineer	
13	Senior Data Engineer	
14	Software Engineer	
15	Armenia	
16	Data Analyst	
17	Data Engineer	
18	Machine Learning Engineer	
19	Australia	
20	Business Analyst	
21	Cloud Engineer	
22	Data Analyst	
23	Data Engineer	
24	Data Scientist	
25	Machine Learning Engineer	
26	Senior Data Analyst	
27	Senior Data Engineer	

Trick to sort in pivot tables:

Be there / keep your active cell on the cell on the basis of which you want to apply filter

The screenshot shows an Excel PivotTable with the following structure:

Row Labels	Salary Year Average	Job Count
Albania		
Algeria		
Data Analyst	₹ 44,100	1
Data Engineer		
Argentina		
Business Analyst	₹ 1	
Cloud Engineer	₹ 1	
Data Analyst	₹ 1	
Data Engineer	₹ 1	
Data Scientist	₹ 1	
Machine Learning Engineer	₹ 1	
Senior Data Engineer	₹ 1	
Software Engineer	₹ 1	
Armenia		
Data Analyst	₹ 1	
Data Engineer	₹ 1	
Machine Learning Engineer	₹ 1	

A context menu is open over the cell containing '₹ 44,100' and '1'. The menu includes options like Copy, Format Cells, Refresh, Delete PivotTable, Sort, Remove "Salary Year Average", Summarize Values By, Show Values As, Show Details, Value Field Settings, and PivotTable Options. The 'Sort' option is expanded, showing 'Sort Smallest to Largest' and 'Sort Largest to Smallest'.

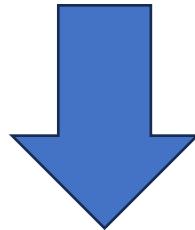
This will sort largest inside each country (But countries won't be sorted...)

The screenshot shows the same Excel PivotTable after sorting by the 'Salary Year Average' field. The data is sorted within each country group, with the highest salary values appearing first.

Row Labels	Salary Year Average	Job Count
Albania	₹ 79,473	5
Algeria	₹ 44,550	2
Data Engineer	₹ 45,000	1
Data Analyst	₹ 44,100	1
Argentina	₹ 1,07,314	34
Cloud Engineer	₹ 1,97,500	1
Software Engineer	₹ 1,87,250	2
Senior Data Engineer	₹ 1,45,667	3
Data Scientist	₹ 99,537	6
Data Engineer	₹ 99,105	14
Machine Learning Engineer	₹ 85,515	2
Data Analyst	₹ 85,436	4
Business Analyst	₹ 71,100	2
Armenia	₹ 84,008	5
Data Analyst	₹ 1,00,500	2
Machine Learning Engineer	₹ 87,021	2
Data Engineer	₹ 45,000	1

Now,

Row Labels	Salary Year				
Albania		₹ 79,473	5		
Algeria					
Data Engineer					
Data Analyst					
Argentina		₹			
Cloud Engineer		₹			
Software Engineer		₹			
Senior Data Engineer		₹			
Data Scientist					
Data Engineer					
Machine Learning Engineer					
Data Analyst					
Business Analyst					
Armenia					
Data Analyst		₹			
Machine Learning Engineer					



	A	B	C	D	E
1	Row Labels	Salary Year Average	Job Count		
2	Belarus	400000	1		
3	Data Analyst	400000	1		
4	Data Scientist				
5	Russia	292500	4		
6	Software Engineer	320000	1		
7	Data Scientist	285000	2		
8	Cloud Engineer	280000	1		
9	Bahamas	₹ 2,01,512	21		
10	Machine Learning Engineer	228750	2		
11	Data Engineer	219405.5	8		
12	Senior Data Engineer	183928.5714	7		
13	Software Engineer	182875	4		
14	Data Analyst				
15	Dominican Republic	157500	1		
16	Senior Data Scientist	157500	1		
17	Namibia	140857.1429	7		
18	Senior Data Engineer	190000	1		

Report Layout:

File					Home	Insert	Draw	Page Layout	Formulas	Data	Review	View	Help	PivotTable Analyze	Design									
Subtotals		Grand Totals		Report Layout		Blank Rows		☒ Row Headers		☐ Banded Rows														
								☒ Column Headers		☐ Banded Columns														
Layout					Options					PivotTable Styles														
D7																								
1					job_count					job_title_short					Salary Year Average					Job Count				
2					Belarus					Data Analyst					400000					1				
3					Belarus Total										400000					1				
4					Russia					Software Engineer					320000					1				
5					Russia					Data Scientist					285000					2				
6					Russia					Cloud Engineer					280000					1				
7					Russia Total										292500					4				
8					Bahamas					Machine Learning Engineer					228750					2				
9					Bahamas					Data Engineer					219405.5					8				
10					Bahamas					Senior Data Engineer					183928.5714					7				
11					Bahamas					Software Engineer					182875					4				
12					Bahamas Total										₹ 2,01,512					21				
13					Dominican Republic					Senior Data Scientist					157500					1				
14					Dominican Republic Total										157500					1				
15					Namibia					Senior Data Engineer					190000					1				
16					Namibia					Business Analyst					145000					1				

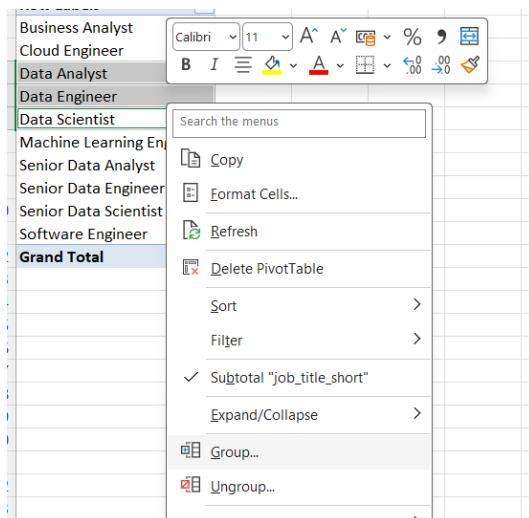
Row Labels	
Jan	
01-Jan	
2023-01-01 00:01:11	
2023-01-01 00:02:56	
2023-01-01 00:06:05	
2023-01-01 00:17:55	
2023-01-01 00:17:58	
2023-01-01 00:18:21	
2023-01-01 00:21:03	
2023-01-01 00:26:50	
2023-01-01 03:11:43	
2023-01-01 04:18:15	
2023-01-01 06:00:26	
2023-01-01 06:04:36	
2023-01-01 06:04:43	
2023-01-01 06:06:43	
2023-01-01 06:08:09	
2023-01-01 06:18:45	
2023-01-01 08:04:48	
2023-01-01 09:17:03	
2023-01-01 14:00:03	
2023-01-01 14:29:27	
2023-01-01 15:01:27	
2023-01-01 15:03:58	
2023-01-01 17:06:41	
2023-01-01 17:12:28	
2023-01-01 21:01:11	
2023-01-01 23:01:14	
2023-01-01 23:04:27	
2023-01-01 23:27:45	
02-Jan	
03-Jan	
04-Jan	
05-Jan	
06-Jan	
07-Jan	
08-Jan	
09-Jan	
10-Jan	
11-Jan	
12-Jan	

Also, Inserted Job Count

	Row Labels	Job Count
1	Jan	3,1
2	Feb	2,5
3	Mar	2,9
4	Apr	2,5
5	May	2,6
6	Jun	3,2
7	Jul	2,8
8	Aug	3,5
9	Sep	2,6
10	Oct	2,5
11	Nov	1,963
12	Dec	2,049
13	Grand Total	32,672
14		
15		
16		

Note: If you want to analyse why some months have low or high job count than others, then you can click like this and can navigate to “Show Details” – This is very cool 🤓 feature.

Manual Grouping



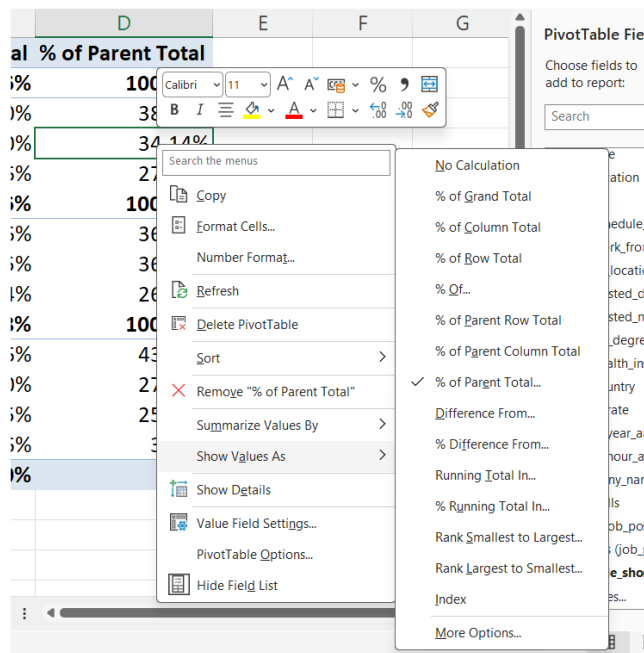
To calculate % of Grand Total, we can do this:

The screenshot shows a PivotTable with the following data:

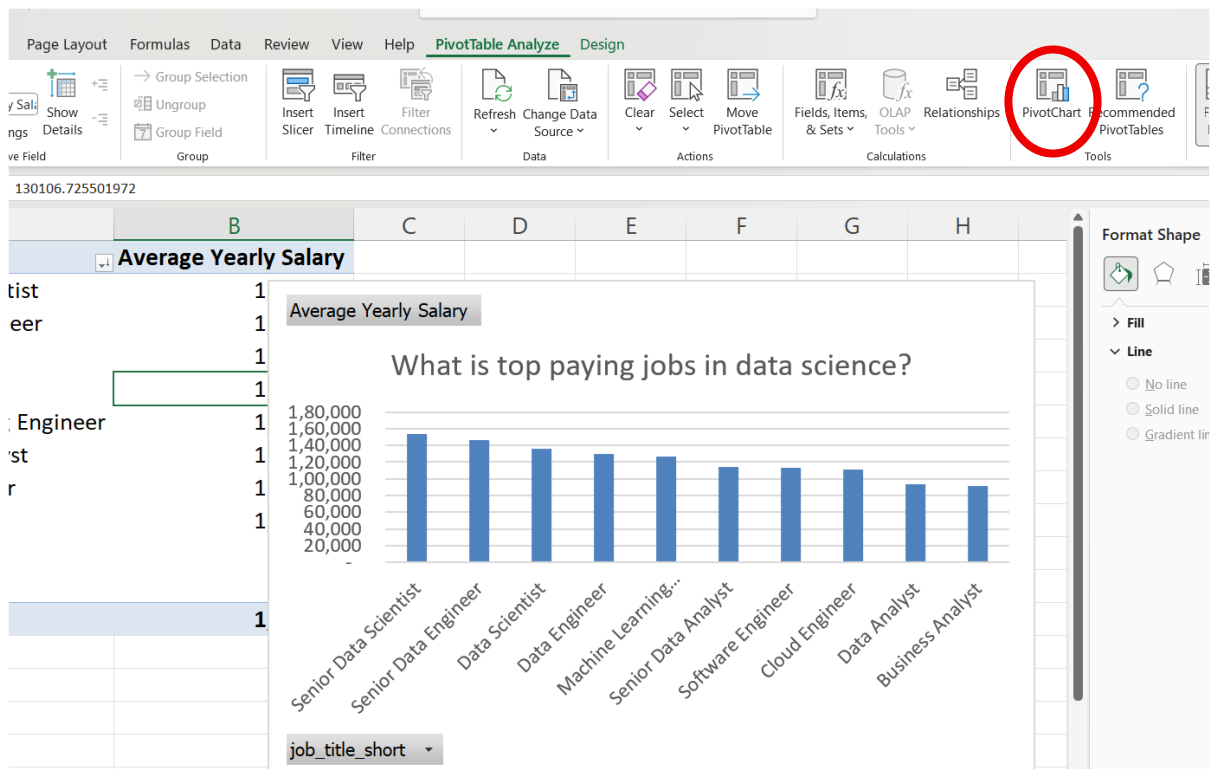
Job Title	% of Grand Total
Business Analyst	76.16%
Cloud Engineer	29.40%
Data Analyst	26.00%
Data Engineer	20.76%
Data Scientist	16.86%
Machine Learning Engineer	6.16%
Senior Data Analyst	6.15%
Senior Data Engineer	4.54%
Senior Data Scientist	6.98%
Software Engineer	3.06%
Grand Total	100.00%

The 'Value Field Settings' dialog box is open, showing the 'Show values as' dropdown set to '% of Grand Total'. The 'PivotTable Fields' task pane is also visible on the right, showing the 'job_title_short' field selected for the 'Values' area.

For % of Parent Total:

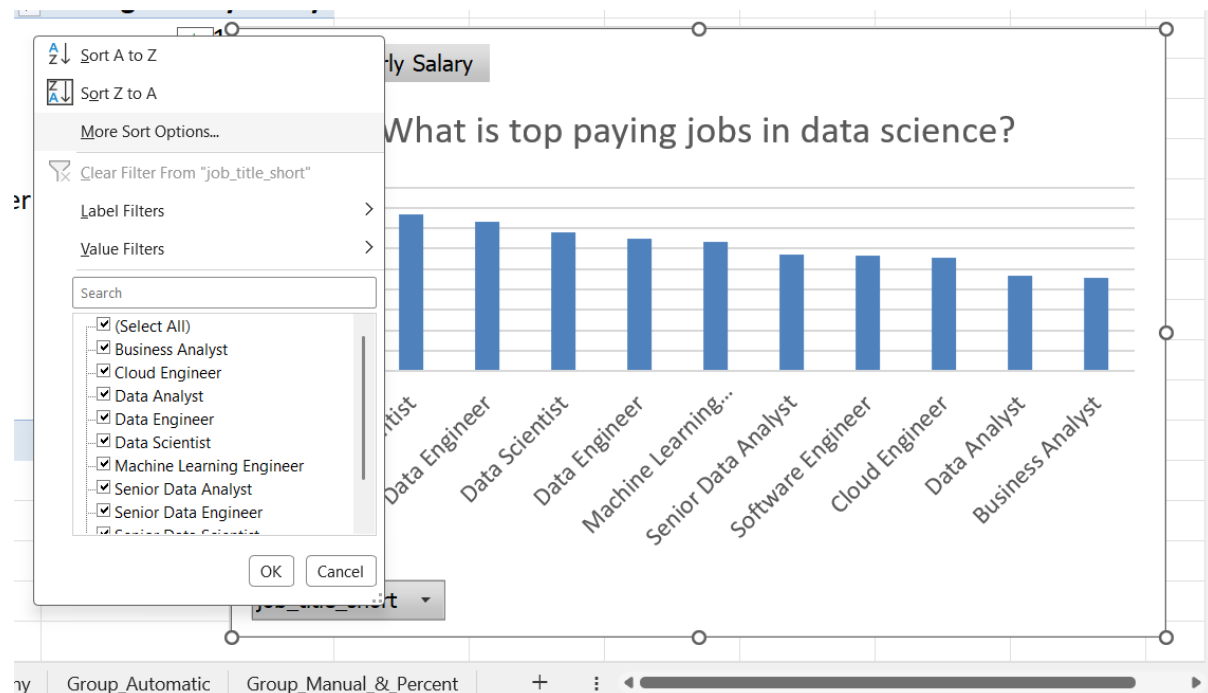


5.3 Pivot Charts

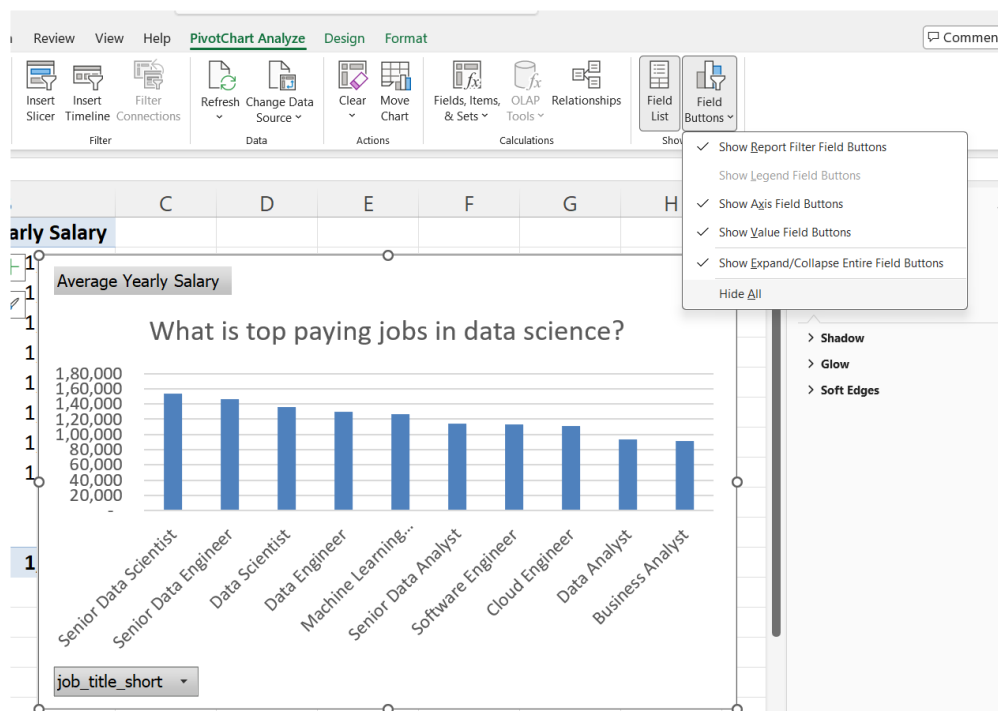


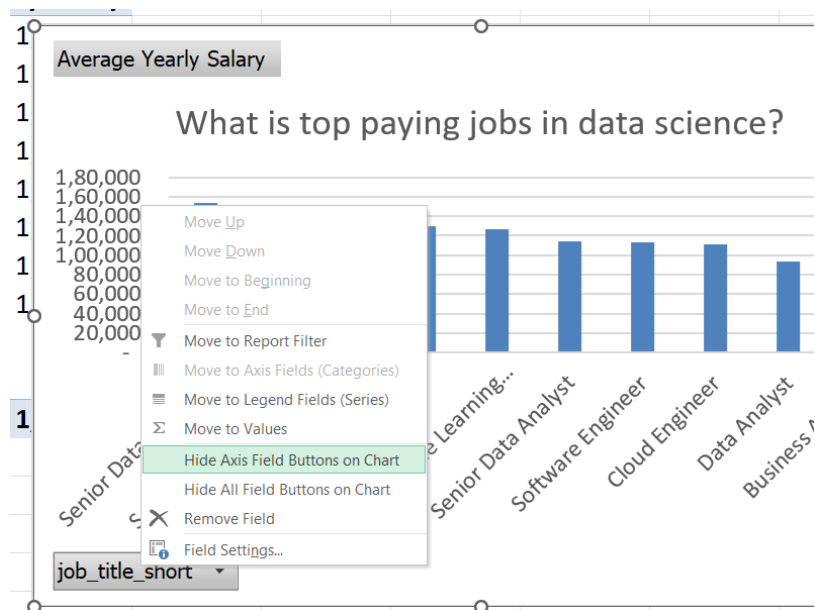
Inserting Pivot Chart.

We can sort chart data here only in case of pivot chart,



We can hide this fields from chart as well





6:15:31